## 6.1. Introduction

We can survive for weeks without food, days without water, but only minutes without air. Thus, we inhale a lungful of air every few seconds, keep some of the molecules for our own end, and some of the molecules that our body no longer needs, and exhale the mixture back into the surrounding air. The air around us is in the gaseous state, which is the simplest of the states of matter. Although the chemical behaviour of gases depends on their composition, all the gases have remarkably similar physical behaviour.

DO YOU KNOW?

Do you know the difference between gas and vapour?

Gas is a substance that is normally in a gaseous state at room temperature and 1 atm pressure, while vapour is the gaseous form of any substance that is a liquid or solid at room temperature and 1 atm pressure.

Earth is surrounded by an atmosphere of air whose composition in volume percentage is roughly 78 % nitrogen, 21 % oxygen and 1 % other gases. Of the known elements, only eleven are gases under normal atmospheric conditions. The elements hydrogen ($H_2$), nitrogen ($N_2$), oxygen ($O_2$), fluorine ($F_2$) and chlorine ($Cl_2$) exist as gaseous diatomic molecules. Another form of oxygen, namely, ozone ($O_3$) is also a gas at room temperature. The noble gases, namely, helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe) and radon (Rn) of 18$^{th}$ group are monatomic gases. Compounds such as carbon monoxide (CO), carbon dioxide ($CO_2$), nitrogen dioxide ($NO_2$) and nitric oxide (NO) also exist in gaseous state under normal atmospheric conditions. In this unit you will learn the properties of gases and other related phenomena.

## 6.2. The Gas Laws

The gas laws have played a major role in the development of chemistry. The physical properties of all gases are governed by the gas laws that were formulated based on the studies of the properties like pressure, volume, etc., as a function of temperature. Before studying the gas laws in detail, let us understand an important parameter, namely, the pressure.

Pressure is defined as force divided by the area to which the force is applied. The SI unit of pressure is pascal which is defined as 1 Newton per square meter ($Nm^{-2}$). There are other units that are commonly used and their relation with the SI unit is as follows.

$$\text{Pressure} = \frac{\text{Force (N or kg m s}^{-2})}{\text{Area (m}^2)}$$

**Table 6.1 Units of pressure**

| Unit | Symbol | Value ($Nm^{-2}$ or $kgm^{-1}s^{-2}$) |
|---|---|---|
| pascal | Pa | 1 Pa = 1 $Nm^{-2}$ |
| atmosphere | atm | 1 atm = 101325 Pa |
| millimeter of mercury | mmHg | 1 mmHg = 133.322 Pa |
| bar | bar | 1 bar = $10^5$ Pa |
| torr | Torr | 1 Torr = 133.322 Pa |
| pound per square inch | psi | 1 psi = 6894.76 pa |



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